Not a finite union of proper subspaces
Artin, 3.7.2

Theorem 1. *Let V be a vector space over an infinite field F. Then V is not the union of finitely many proper subspaces.*

Proof. We prove the following stronger assertion: for every finite set $\mathcal{S}$ of proper subspaces of V, there exists a vector $v \in V$ such that $v \notin W$ for every $W \in \mathcal{S}$. In particular, no finite collection of proper subspaces can have union equal to V.

We argue by induction on the finite set $\mathcal{S}$.

If $\mathcal{S} = \varnothing$, then $v = 0$ satisfies the requirement vacuously.

For the inductive step, suppose the assertion is true for a finite set $\mathcal{S}$ of proper subspaces, and consider $\mathcal{S} \cup \{A\}$, where A is a proper subspace not in $\mathcal{S}$. By the induction hypothesis applied to $\mathcal{S}$, there exists $x \in V$ such that

$$x \notin U \qquad \text{for every } U \in \mathcal{S}.$$

We distinguish two cases.

Case 1: $x \in A$**.**

Since A is proper, $A \neq V$, so there exists $y \in V$ with $y \notin A$.

We first observe that for any subspace U with $x \notin U$, there is at most one scalar $a \in F$ such that $y + ax \in U$. Indeed, suppose that $y + ax \in U$ and $y + bx \in U$ for scalars $a, b \in F$. Since U is a subspace,

$$(a - b)x = (y + ax) - (y + bx) \in U.$$

If $a \neq b$, then $a - b \neq 0$, and hence

$$x = (a - b)^{-1}(a - b)x \in U,$$

contradicting $x \notin U$. Thus $a = b$.

Now, for each $U \in \mathcal{S}$, if there exists a scalar $d \in F$ such that $y + dx \in U$, choose one such scalar and denote it by d_U; if no such scalar exists, set $d_U = 0$. The set

$$D = \{d_U : U \in \mathcal{S}\}$$

is finite. Since F is infinite, we may choose $c \in F$ with $c \notin D$.

We claim that $y + cx$ lies in no member of $\mathcal{S} \cup \{A\}$. First, suppose for some $U \in \mathcal{S}$ that $y + cx \in U$. Then such a scalar as in the definition of d_U does exist, namely c, so the chosen scalar d_U satisfies $y + d_U x \in U$. By the uniqueness observed above, applied to U (which satisfies $x \notin U$), we conclude that $d_U = c$. Hence $c \in D$, a contradiction. Therefore $y + cx \notin U$ for every $U \in \mathcal{S}$.

Next, $y + cx \notin A$: indeed, since $x \in A$, we have $cx \in A$; if also $y + cx \in A$, then

$$y = (y + cx) - cx \in A,$$

contradicting the choice of y.

Thus $v = y + cx$ lies in no member of $\mathcal{S} \cup \{A\}$.

Case 2: $x \notin A$**.**

Then x itself lies in no member of $\mathcal{S} \cup \{A\}$: it lies in no member of $\mathcal{S}$ by the induction hypothesis, and it does not lie in A by assumption of this case.

This completes the inductive step, and hence the proof of the stronger assertion. Consequently, V cannot be the union of finitely many proper subspaces. $\square$